417521 — MACHINE LEARNING

UNIT II: REGRESSION

DETAILED • EASY TO UNDERSTAND • EXAM-FOCUSED

Fourth Year AI & Data Science (AIDS) • SPPU 2020 Course

These notes follow the Unit II scope in the syllabus you shared: Regression and Need, Regression vs Correlation, regression types, Bias–Variance, Overfitting/Underfitting, Polynomial, Stepwise, Decision Tree, Random Forest, Support Vector, Ridge, Lasso, ElasticNet and Bayesian Linear Regression, MSE, MAE, RMSE, R^2, Adjusted R^2, and the sales-prediction case study.

Regression — Basic Idea

REMEMBER: Regression = learn a relationship from data → predict a numerical value.

1. UNIT II — ZERO-SKIP CHECKLIST

The official syllabus is the boundary. Every point below is covered in explanatory form; later-unit classification/clustering and unrelated advanced topics are intentionally excluded.

Syllabus point	Coverage
Introduction + Need	Definition, intuition, need and examples
Regression vs Correlation	Detailed comparison
Univariate vs Multivariate	Definition + comparison
Linear vs Nonlinear	Definition + comparison
Simple vs Multiple Linear	Equations + detailed comparison
Bias–Variance	Concept + tradeoff
Overfitting + Underfitting	Causes, signs, comparison
Polynomial + Stepwise	Concept + working
Decision Tree + Random Forest	Working + comparison
Support Vector Regression	ε-insensitive concept
Ridge + Lasso + ElasticNet	Regularization + comparison
Bayesian Linear	Prior, likelihood, posterior
MSE, MAE, RMSE	Formula + interpretation + numerical
R^2 + Adjusted R^2	Formula + interpretation
Sales case study	Ridge/Lasso/ElasticNet comparison

EXAM POINT: Do not skip the short comparison topics; they are direct exam questions.

2. INTRODUCTION TO REGRESSION

Regression is a supervised Machine Learning technique used when the target/output is numerical. The model learns a relationship from existing observations and uses that learned relationship to estimate the numerical target for new observations.

For example, historical advertising expenditure and sales can be used to learn how sales vary with expenditure and then estimate sales for a new expenditure value.

Regression — Basic Idea

INPUT FEATURES → **REGRESSION MODEL** → **NUMERICAL PREDICTION**

- Y = dependent/target numerical variable.
- X = independent/predictor variable(s).
- $\hat{y}$ = predicted value.
- Residual = actual value − predicted value.
- Regression is useful for modelling a relationship and making numerical predictions.

EXAM POINT: Define regression, identify X and Y, give an example and draw the model relationship.

REMEMBER: Regression = numerical prediction.

3. NEED OF REGRESSION

Regression is needed when the output is a quantity that can take numerical values. Historical data often contains patterns that are useful for estimating future or unknown values.

A regression model converts these patterns into a learned relationship, allowing numerical prediction and supporting practical decisions.

- Sales prediction and demand forecasting.
- House/property price estimation.
- Salary estimation.
- Inventory or demand planning.
- Prediction of numerical measurements.
- Understanding how predictors are associated with a numerical target.
- Supporting planning, budgeting and decision-making.

EXAM POINT: Explain at least five needs and one practical example.

REMEMBER: Need = predict + estimate + forecast + understand + decide.

4. REGRESSION VS CORRELATION

Correlation measures the strength and direction of association between variables. Regression builds a model in which predictors are used to estimate a numerical target. Therefore, although the two concepts are related, they are not interchangeable.

Basis	Correlation	Regression
Purpose	Measure association	Build predictive/estimating model
Direction	Association is symmetric	Predictor → target is specified
Output	Correlation coefficient	Regression model/equation + prediction
Prediction	Not primary purpose	Central purpose
Example	Strength of relation between hours and marks	Predict marks from hours

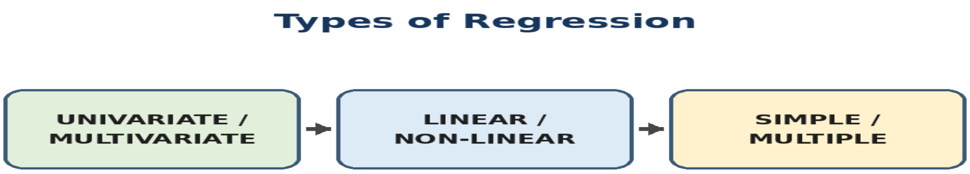
EXAM POINT: Write this as a table for a comparison question.

REMEMBER: Correlation = association; Regression = prediction/model.

5. UNIVARIATE VS MULTIVARIATE REGRESSION

The syllabus specifically asks for Univariate versus Multivariate Regression. The main basis is the number of dependent/output variables being modelled.

Do not confuse this with multiple regression: multiple linear regression generally means multiple predictors are used for one numerical target.



Point	Univariate	Multivariate
Outputs	One	More than one
Focus	One outcome	Multiple outcomes
Memory	Uni = one	Multi = many

- Univariate: one dependent/output variable is modelled.
- Multivariate: more than one dependent/output variable is modelled.
- The distinction is about outputs, not simply the number of input predictors.

EXAM POINT: State the basis first: number of dependent/output variables.

REMEMBER: Univariate = one output; Multivariate = multiple outputs.

6. LINEAR VS NONLINEAR REGRESSION

Linear regression uses a linear model form. Nonlinear regression uses a nonlinear model form when the chosen linear form cannot adequately represent the relationship.

At syllabus level, remember the distinction by the form of the model rather than memorising only the appearance of a graph.

Basis	Linear	Nonlinear
Model form	Linear	Nonlinear
Relationship intuition	Straight-line/basic linear form	Curved/other nonlinear form
Complexity	Usually lower	Can be higher
Risk	May underfit curves	May overfit if excessively flexible

- Linear models are generally simpler to interpret.
- Nonlinear models can represent more complex/curved relationships.
- Too little flexibility may underfit; excessive flexibility can overfit.

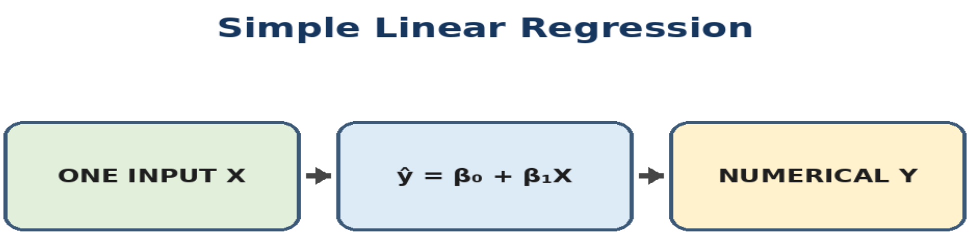
EXAM POINT: Give the model-form distinction and one example.

REMEMBER: Linear = linear model form; nonlinear = nonlinear form.

7. SIMPLE LINEAR REGRESSION

Simple Linear Regression predicts one numerical target using exactly one predictor. It is the basic linear regression model and is important for understanding the intercept, slope and prediction.

$$\hat{y} = \beta_0 + \beta_1 X$$



- β_0 = intercept.
- β_1 = slope/coefficient.
- X = the single predictor.
- $\hat{y}$ = predicted numerical target.
- Example: predict marks from study hours.

EXAM POINT: Explain every symbol, draw the fitted line and give one simple example.

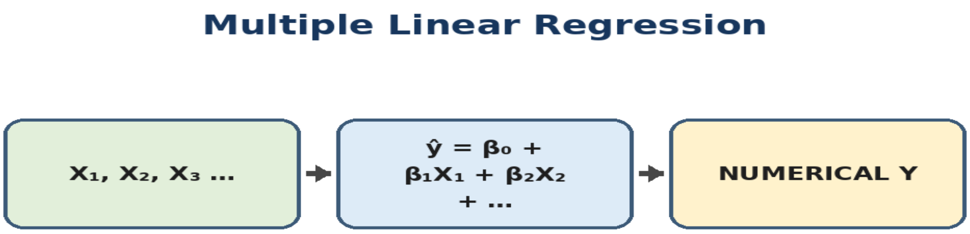
REMEMBER: Simple = exactly one predictor.

8. MULTIPLE LINEAR REGRESSION

Multiple Linear Regression predicts one numerical target using two or more predictors. It is useful when several factors jointly contribute to the target.

The coefficient of each predictor describes its contribution in the fitted linear model while the other included predictors are held fixed in the usual interpretation.

$$\hat{y} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p$$



- β_0 = intercept.
- $X_1 \dots X_p$ = predictors.
- $\beta_1 \dots \beta_p$ = coefficients.
- $\hat{y}$ = predicted target.
- Example: predict sales from advertising expenditure, traffic and another selected feature.

▯ EXAM POINT: Remember: multiple predictors, one numerical target.

▯ REMEMBER: Multiple = many X → one Y.

9. SIMPLE VS MULTIPLE LINEAR REGRESSION

Both are linear regression models. Their main difference is the number of predictors.

Basis	Simple	Multiple
Predictors	Exactly 1	2 or more
Target	1 numerical	1 numerical
Equation	$\beta_0 + \beta_1 X$	$\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$
Complexity	Lower	Higher
Example	Marks ← study hours	Sales ← several features

▯ EXAM POINT: Use a table and include the equations.

▯ REMEMBER: Simple = 1 X; Multiple = many X.

10. RESIDUAL / PREDICTION ERROR

A fitted regression model produces a prediction, but the prediction may differ from the actual value. The residual measures this difference and is the foundation for MSE, MAE and RMSE.

$$e_i = y_i - \hat{y}_i$$

- Positive residual: actual > prediction.
- Negative residual: predicted > actual.
- Zero residual: exact prediction.
- Large absolute residual: prediction is far from actual.
- MSE squares residuals; MAE takes absolute residuals; RMSE is √MSE.

▯ EXAM POINT: This concept is essential when explaining the later evaluation metrics.

▯ REMEMBER: Actual − predicted = residual.

11. BIAS–VARIANCE TRADEOFF

Bias is error caused by overly simple assumptions, while variance is associated with excessive sensitivity to the particular training data. A useful model needs a balance between the two.

As model complexity increases, bias often decreases because the model can represent richer patterns, but variance can increase because the model may start fitting training-specific noise.

Bias–Variance Tradeoff

- High bias → important relationships may be missed.
- High variance → noise or accidental training patterns may be learned.
- Very simple model → underfitting tendency.
- Very complex model → overfitting tendency.
- Goal → good generalization on unseen observations.

□ **EXAM POINT: Define bias, define variance, explain model-complexity effect and connect both to underfitting and overfitting.**

□ **REMEMBER: High bias → underfit; high variance → overfit.**

12. UNDERFITTING AND OVERFITTING

Underfitting occurs when the model is too simple to capture the important pattern. Overfitting occurs when the model fits the training data too specifically and learns noise or accidental details that do not generalize.

Underfitting — Good Fit — Overfitting

Property	Underfit	Good fit	Overfit
Complexity	Too low	Balanced	Too high
Training error	High	Low/appropriate	Very low
Unseen error	High	Low/appropriate	High
Main issue	Missed pattern	Generalizes well	Learns noise

- Underfitting: high training error and high unseen/test error.
- Overfitting: very low training error but high unseen/test error.
- Underfitting is associated with high bias.
- Overfitting is associated with high variance.
- Suitable model complexity aims for good generalization.

□ **EXAM POINT: Explain both training and unseen/test behaviour, not just the definitions.**

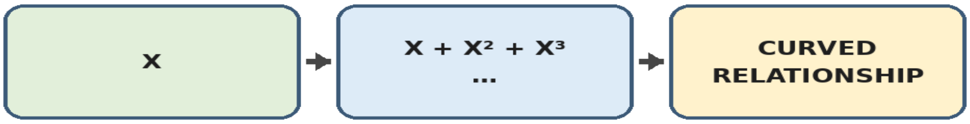
□ **REMEMBER: Underfit = not learned enough; overfit = learned too specifically.**

13. POLYNOMIAL REGRESSION

Polynomial Regression introduces powers of a predictor so that the fitted model can represent a curved relationship. The degree determines how flexible the polynomial can be.

$$\hat{y} = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \dots + \beta_n X^n$$

Polynomial Regression



- Degree 1 gives a basic linear form.
- X^2 can represent quadratic curvature.
- Higher powers allow more flexible shapes.
- Workflow: choose degree → create polynomial terms → fit → evaluate.
- Very high degree can overfit.

▯ **EXAM POINT: Explain why powers of X allow a curve and why high degree can overfit.**

▯ **REMEMBER: Polynomial = powers of X → curve.**

14. STEPWISE REGRESSION

Stepwise Regression is a predictor-selection procedure. Variables are added or removed step by step using a chosen criterion.

Method	Starting point	Action
Forward	Small/empty model	Add
Backward	Full candidate model	Remove
Stepwise	Flexible	Add + remove

- Forward Selection: start with no/small set and add useful predictors.
- Backward Elimination: start with all candidates and remove predictors that fail the criterion.
- Stepwise Selection: permits addition and removal during the procedure.
- Continue until the stopping rule is reached.
- Purpose: obtain a useful subset of predictors.

▯ **EXAM POINT: Explain Forward, Backward and Stepwise approaches separately.**

▯ **REMEMBER: Forward = add; Backward = remove; Stepwise = both.**

15. DECISION TREE REGRESSION

Decision Tree Regression predicts a numerical target by repeatedly splitting observations according to feature-based decisions. A path through the tree eventually reaches a leaf that provides a numerical prediction.

Tree and Random Forest Regression

- Root node = first split.
- Internal node = later decision.
- Branch = outcome of a split.
- Leaf = final numerical prediction.
- Can represent nonlinear patterns and feature interactions.
- Very deep trees can overfit.

▯ **EXAM POINT: Draw and label Root, Internal Node, Branch and Leaf.**

▯ **REMEMBER: Decision Tree = split → split → leaf.**

16. RANDOM FOREST REGRESSION

Random Forest Regression is an ensemble method that combines many decision-tree regressors. Randomness makes the trees diverse and their numerical predictions are aggregated, commonly by averaging.

Decision Tree	Random Forest
One tree	Many trees
One structure	Many structures combined
Can be unstable if very deep	Aggregation can improve stability

- Build multiple trees.
- Introduce randomness through sampling and/or feature selection.
- Each tree produces a numerical prediction.
- Aggregate predictions, commonly by averaging.
- Combining trees can reduce the variance/instability of a single tree.
- Can model complex nonlinear relationships.
- Less interpretable and more computationally demanding than one small tree.

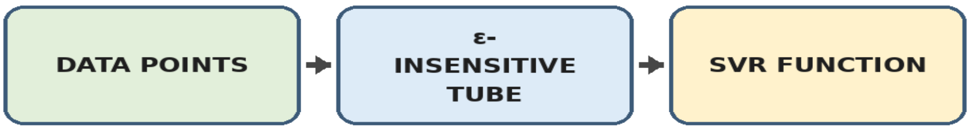
□ **EXAM POINT: Main phrase: many trees + aggregation.**

□ **REMEMBER: Random Forest = many trees → aggregate.**

17. SUPPORT VECTOR REGRESSION (SVR)

Support Vector Regression adapts the support-vector idea to regression. It creates an ε-insensitive region around the regression function. Small deviations within this tolerance are treated differently from deviations outside it.

Support Vector Regression



- ε defines the tolerance width.
- The ε-insensitive tube surrounds the regression function.
- Support vectors are influential observations that help determine the fitted function/boundaries.
- Kernels can support nonlinear regression.
- Parameter and kernel choices affect the model.

□ **EXAM POINT: Draw the ε tube and label regression function, boundaries and support vectors.**

□ **REMEMBER: SVR = ε-tube + support vectors.**

18. RIDGE REGRESSION

Ridge Regression is linear regression with an L2 regularization penalty. The penalty discourages large coefficient values and helps control model complexity.

$$\textbf{Objective} \approx \Sigma(y_i - \hat{y}_i)^2 + \lambda\Sigma\beta_j^2$$

Regularization

- L2 means the penalty uses squared coefficients.
- λ controls regularization strength.
- Larger λ generally gives stronger shrinkage.

- Ridge normally shrinks coefficients toward zero rather than forcing them exactly to zero.
- Useful when predictors are correlated.
- Can reduce overfitting tendency.

□ EXAM POINT: Explain L2 and λ and their effect on coefficients.

□ REMEMBER: Ridge = L2 = shrink.

19. LASSO REGRESSION

Lasso Regression uses an L1 regularization penalty based on absolute coefficient values. A key property is that some coefficients can become exactly zero, giving Lasso a feature-selection effect.

Objective $\approx \Sigma(y_i - \hat{y}_i)^2 + \lambda\Sigma|\beta_j|$

- L1 = absolute coefficient penalty.
- λ controls regularization strength.
- Some coefficients may become exactly zero.
- Zero coefficients remove those predictors from the fitted linear model.
- Therefore Lasso can create a sparse, simpler model.
- Selection among correlated predictors can be unstable.

□ EXAM POINT: Use the phrase: L1 penalty can make coefficients exactly zero.

□ REMEMBER: Lasso = L1 = select.

20. ELASTIC NET REGRESSION

Elastic Net combines L1 and L2 regularization. The L1 component can encourage sparsity/feature selection, while the L2 component encourages coefficient shrinkage.

Conceptually: loss + $\lambda_1\Sigma|\beta_j| + \lambda_2\Sigma\beta_j^2$

- Uses absolute-value and squared-coefficient penalties.
- Can set some coefficients to zero.
- Also shrinks coefficients.
- Useful when predictors are numerous and/or correlated.
- Combines useful properties of Lasso and Ridge.

□ EXAM POINT: Compare Elastic Net with Ridge and Lasso.

□ REMEMBER: Elastic Net = L1 + L2.

21. RIDGE vs LASSO vs ELASTIC NET

These three methods are central to the Unit II case study. Their main difference is the regularization penalty and resulting coefficient behaviour.

Basis	Ridge	Lasso	Elastic Net
Penalty	L2	L1	L1 + L2
Effect	Shrink	Shrink + can zero	Shrink + can zero

Basis	Ridge	Lasso	Elastic Net
Feature selection	Not primary	Yes	Yes
Memory	SHRINK	SELECT	COMBINE

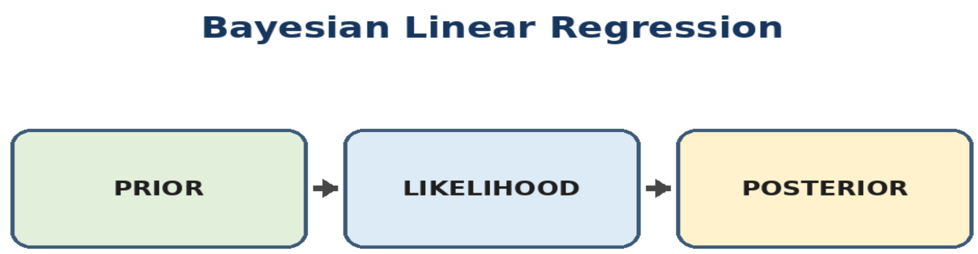
EXAM POINT: Very important direct comparison question.

REMEMBER: Ridge shrinks; Lasso selects; Elastic Net combines.

22. BAYESIAN LINEAR REGRESSION

Bayesian Linear Regression treats regression parameters probabilistically. It combines prior information with evidence from observed data to produce a posterior distribution.

$$\textbf{Posterior} \propto \textbf{Likelihood} \times \textbf{Prior}$$



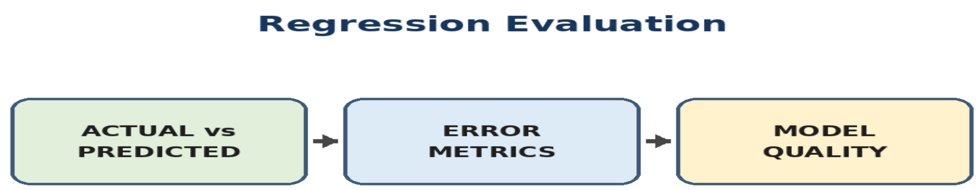
- Prior = information/belief about parameters before current data.
- Likelihood = compatibility of observed data with possible parameter values.
- Posterior = updated parameter distribution after considering the data.
- Posterior represents updated uncertainty about parameters.
- Useful when uncertainty needs to be represented explicitly.

EXAM POINT: Define Prior, Likelihood and Posterior and show the relationship.

REMEMBER: Prior + evidence → Posterior.

23. EVALUATION METRICS — WHY THEY ARE NEEDED

After training a regression model, predictions must be evaluated. The syllabus specifically includes MSE, MAE, RMSE, R-squared and Adjusted R-squared.



Metric	Type	General direction
MSE	Prediction error	Lower
MAE	Prediction error	Lower
RMSE	Prediction error	Lower
R^2	Explained variation	Higher
Adjusted R^2	Explained variation + complexity adjustment	Higher

EXAM POINT: Know which metrics are error measures and which are explained-variation measures.

REMEMBER: MSE/MAE/RMSE = errors; R^2/Adjusted R^2 = explained variation.

24. MEAN SQUARED ERROR (MSE)

MSE is the average of squared prediction errors. Because the residual is squared, large errors contribute much more strongly to the final value.

$$\mathbf{MSE = (1/n)\ \Sigma(y_i - \hat{y}_i)^2}$$

- n = number of observations.

- y_i = actual target; $\hat{y}_i$ = predicted target.
- MSE = 0 means perfect prediction.
- Lower MSE is generally better for the same evaluation setting.
- MSE has squared target units.
- Large errors receive a strong penalty.

□ **EXAM POINT: Explain why squaring matters, not just the formula.**

□ **REMEMBER: MSE = square residuals → average.**

25. MEAN ABSOLUTE ERROR (MAE)

MAE is the average absolute difference between actual and predicted values. It represents the average magnitude of prediction error.

$$\mathrm{MAE} = (1/n)\,\Sigma|y_i - \hat{y}_i|$$

- Absolute value prevents positive and negative errors from cancelling.
- MAE = 0 means perfect prediction.
- Lower MAE is generally better.
- MAE is in the same units as the target.
- Very large errors are not given the same squared penalty as in MSE.

□ **EXAM POINT: Explain formula + units + interpretation.**

□ **REMEMBER: MAE = absolute residuals → average.**

26. ROOT MEAN SQUARED ERROR (RMSE)

RMSE is the square root of MSE. It keeps the sensitivity to large errors created by squaring, but returns the measure to the original target units.

$$\mathrm{RMSE} = \sqrt{\mathrm{MSE}}$$

- RMSE = 0 means perfect prediction.
- Lower is generally better.
- Large errors strongly affect RMSE.
- Same units as the target.
- Often easier to interpret than MSE because it uses target units.

□ **EXAM POINT: Explain why RMSE is more directly interpretable than MSE.**

□ **REMEMBER: RMSE = √MSE.**

27. R-SQUARED (R^2)

R^2 measures variation in the dependent variable explained by the fitted model relative to a mean-based baseline.

$$R^2 = 1 - [\Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2]$$

- y_i = actual value.
- $\hat{y}_i$ = predicted value.
- $\bar{y}$ = mean target value.
- Numerator = residual/squared unexplained variation.
- Denominator = total variation around the target mean.
- Higher R^2 generally indicates more explained variation relative to the baseline.
- R^2 alone does not guarantee good unseen-data performance.

□ **EXAM POINT: Do not call R^2 simply 'accuracy'; explain explained variation.**

□ **REMEMBER: R^2 = relative explained variation.**

28. ADJUSTED R-SQUARED

Adjusted R^2 modifies R^2 by accounting for the number of predictors and the sample size. It is useful when comparing multiple regression models with different numbers of predictors.

$$\text{Adjusted } R^2 = 1 - [(1 - R^2)(n - 1)/(n - p - 1)]$$

- n = number of observations.
- p = number of predictors.
- It discourages adding predictors that do not provide enough improvement.
- Adjusted R^2 can decrease when an unnecessary predictor is added.
- Useful for comparing models with different predictor counts.

□ **EXAM POINT: Main difference: Adjusted R^2 includes a complexity adjustment.**

□ **REMEMBER: Adjusted R^2 = R^2 with predictor-count adjustment.**

29. COMPLETE METRIC COMPARISON

These metrics are not interchangeable. MSE, MAE and RMSE directly quantify prediction error, whereas R^2 and Adjusted R^2 provide a different view based on explained variation.

Metric	Core idea	Units / interpretation	Direction
MSE	Average squared error	Squared target units	Lower
MAE	Average absolute error	Target units	Lower
RMSE	√MSE	Target units	Lower
R^2	Relative explained variation	Dimensionless	Higher
Adjusted R^2	R^2 adjusted for predictors	Dimensionless	Higher

EXAM POINT: For a full comparison include purpose, formula idea, units and direction.

REMEMBER: Errors: lower. Explained variation: higher.

30. NUMERICAL EXAMPLE — MSE, MAE AND RMSE

Given actual values y = [3, 5, 7] and predicted values ŷ = [2, 5, 8]. First calculate the residuals.

Obs.	Actual	Predicted	Residual	|Residual|	Residual^2
1	3	2	1	1	1
2	5	5	0	0	0
3	7	8	−1	1	1

- MSE = (1 + 0 + 1) / 3 = 0.667.
- MAE = (1 + 0 + 1) / 3 = 0.667.
- RMSE = √0.667 ≈ 0.816.

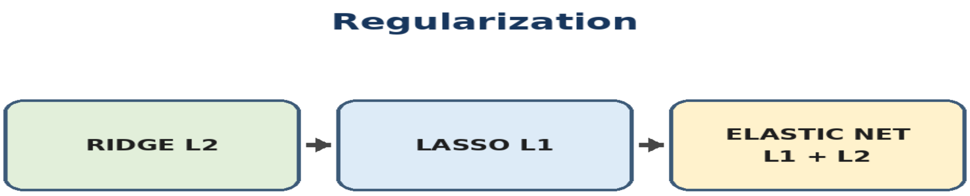
EXAM POINT: Numerical format: Given → residual table → formula → substitution → result → interpretation.

REMEMBER: Show the error calculation before the final metric.

31. SPPU CASE STUDY — SALES PREDICTION

The official Unit II exemplar/case study asks students to build and compare Lasso, Ridge and Elastic Net regression models using selected features to predict sales.

The purpose is to compare regularized regression approaches on the same prediction problem and evaluate their predictive behaviour.



- 1 Define the numerical target as sales.
- 2 Select relevant features.
- 3 Prepare the data for modelling.
- 4 Train Ridge Regression.
- 5 Train Lasso Regression.
- 6 Train Elastic Net Regression.
- 7 Generate predictions.
- 8 Evaluate using suitable regression metrics.
- 9 Compare results and justify the model choice.

EXAM POINT: Keep the answer tied to sales prediction; do not invent numerical results not given in the question.

REMEMBER: Sales → features → Ridge/Lasso/Elastic Net → predictions → metrics → comparison.

32. CASE STUDY — HOW TO COMPARE THE THREE MODELS

A good case-study answer explains what the comparison means rather than merely listing model names.

Question	What to discuss
Which predicts best?	Compare suitable error metric(s); lower error is generally preferred.
Which can simplify the model?	Lasso/Elastic Net can produce zero coefficients.
Why Ridge?	Coefficient shrinkage can be useful, especially with correlated predictors.
Why Elastic Net?	Combines L1 selection behaviour with L2 shrinkage.

□ **EXAM POINT: Explain model building + prediction + evaluation + comparison, not just definitions.**

□ **REMEMBER: Case study = problem → models → predictions → metrics → comparison.**

33. MASTER TABLE — ALL UNIT II REGRESSION TECHNIQUES

Use this after learning the detailed sections; it is a revision aid, not a replacement for full answers.

Technique	Core idea	Memory
Polynomial	Powers of predictors	Curve
Stepwise	Add/remove variables	Selection
Decision Tree	Recursive splits	Split → leaf
Random Forest	Many trees + aggregation	Many trees
SVR	ε-insensitive tube	ε + support vectors
Ridge	L2 penalty	Shrink
Lasso	L1 penalty	Select
Elastic Net	L1 + L2	Combine
Bayesian Linear	Prior + likelihood → posterior	Uncertainty

□ **REMEMBER: Do not write only the memory word in the exam; expand it into a complete explanation.**

34. ADVANTAGES & LIMITATIONS — EXAM READY

Use these as supporting points in long answers.

Technique	Advantage	Limitation
Polynomial	Models curvature	High degree can overfit
Stepwise	Can select a useful subset	Depends on criterion/data
Decision Tree	Nonlinear and visual	Can overfit
Random Forest	Ensemble can reduce variance	Less interpretable
SVR	Flexible; kernels support nonlinear patterns	Parameter/kernel choice matters
Ridge	Shrinkage; useful with correlated predictors	Usually does not zero coefficients
Lasso	Feature selection	Correlated predictors can complicate selection
Elastic Net	Selection + shrinkage	More tuning
Bayesian	Represents uncertainty	Requires probabilistic assumptions

□ **EXAM POINT: Only use limitations relevant to the technique asked; avoid off-syllabus discussion.**

35. □ IMPORTANT QUESTIONS — PRIORITY A

Prepare these first:

- 1. Define Regression and explain its need.
- 2. Differentiate Regression and Correlation.

- 3. Explain Univariate vs Multivariate.
- 4. Differentiate Linear vs Nonlinear.
- 5. Explain Simple Linear Regression with equation.
- 6. Explain Multiple Linear Regression with equation.
- 7. Differentiate Simple and Multiple Linear Regression.
- 8. Explain Bias–Variance Tradeoff.
- 9. Explain Overfitting and Underfitting with diagram.
- 10. Explain Polynomial Regression.
- 11. Explain Stepwise Regression and its approaches.
- 12. Explain Decision Tree Regression.
- 13. Explain Random Forest Regression.
- 14. Explain Support Vector Regression.

EXAM POINT: Technique answer format: Definition → diagram → concept/formula → working → example → advantages/limitations.

36. IMPORTANT QUESTIONS — PRIORITY B

Then prepare these carefully:

- 15. Explain Ridge Regression with L2 regularization.
- 16. Explain Lasso Regression with L1 regularization.
- 17. Explain Elastic Net Regression.
- 18. Differentiate Ridge, Lasso and Elastic Net.
- 19. Explain Bayesian Linear Regression using Prior, Likelihood and Posterior.
- 20. Explain MSE, MAE and RMSE with formulas.
- 21. Explain R^2 with formula and interpretation.
- 22. Explain Adjusted R^2 and its importance.
- 23. Compare all regression evaluation metrics.
- 24. Explain the Ridge/Lasso/Elastic Net sales-prediction case study.

EXAM POINT: Numerical format: Given → Formula → Substitution → Calculation → Result → Interpretation.

REMEMBER: Do not skip formulas in metric answers.

37. HOW TO WRITE A HIGH-QUALITY 5–6 MARK SPPU ANSWER

A good answer should be complete without becoming repetitive. Use a fixed structure so you do not forget important parts.

Question type	Recommended structure
Comparison	Basis → table → conclusion
Technique	Definition → diagram → formula/concept → working → example → pros/cons
Numerical	Given → formula → substitution → calculation → result → interpretation
Case study	Problem → data/features → models → predictions → metrics → comparison

- ✍ Definition — 2–4 clear lines.
- Purpose/Need — explain why it is used.
- Diagram — use a labelled pictorial diagram wherever appropriate.
- Formula/Equation — write it and explain symbols.
- Working/Steps — numbered and logical.
- Example — one easy real-world example.
- Advantages — 3–4 relevant points.
- ⚠ Limitations — 1–3 relevant points.
- Conclusion — one short concluding line.

REMEMBER: Fill pages with explanation, diagrams, formulas and examples—not repeated sentences.

38. MASTER MEMORY CHART

Use this only after understanding the detailed sections.

- Regression → numerical prediction.
- Correlation → association.
- Simple → 1 predictor.
- Multiple → 2+ predictors.

- Univariate → 1 output.
- Multivariate → multiple outputs.
- Linear → linear model form.
- Nonlinear → nonlinear form.
- Bias → overly simple assumptions.
- Variance → sensitivity to training data.
- Underfit → high bias.
- Overfit → high variance.
- Polynomial → powers of X.
- Stepwise → add/remove variables.
- Decision Tree → split → leaf.
- Random Forest → many trees → aggregate.
- SVR → ε-tube + support vectors.
- Ridge → L2 → shrink.
- Lasso → L1 → select.
- Elastic Net → L1 + L2.
- Bayesian → prior + likelihood → posterior.
- MSE/MAE/RMSE → errors.
- R^2/Adjusted R^2 → explained variation.

REMEMBER: Memory charts are for revision; full answers require explanation.

39. ZERO-SKIP FINAL CHECKLIST

The official syllabus confirms the Unit II topic boundary.

- ☐ Regression + Need of Regression
- ☐ Regression vs Correlation
- ☐ Univariate vs Multivariate
- ☐ Linear vs Nonlinear
- ☐ Simple vs Multiple Linear
- ☐ Bias–Variance
- ☐ Overfitting
- ☐ Underfitting
- ☐ Polynomial Regression
- ☐ Stepwise Regression
- ☐ Decision Tree Regression
- ☐ Random Forest Regression
- ☐ Support Vector Regression
- ☐ Ridge Regression
- ☐ Lasso Regression
- ☐ ElasticNet Regression
- ☐ Bayesian Linear Regression
- ☐ MSE
- ☐ MAE
- ☐ RMSE
- ☐ R-squared
- ☐ Adjusted R-squared
- ☐ MSE/MAE/RMSE numerical
- ☐ Ridge vs Lasso vs Elastic Net
- ☐ Sales case study
- ☐ Important questions + answer-writing format

EXAM POINT: If any checkbox is weak, revise that section before moving to the next unit.

REMEMBER: Complete syllabus coverage is more important than memorising the memory chart.